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USE OF SATELLITE DATA FOR CLIMATE CHANGE ADAPTATION AND RESILIENCE STRATEGIES

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Abstract

Around the world, climate change is causing various problems such as rising sea levels, extreme weather, reduction of glaciers and ice caps and ecosystem change. Climate change adaptation is the process of adjusting to the effects of climate change, while climate change resilience is the ability to prepare for and recover from those effects. Building resilience requires the consideration of multiple aspects of environmental, social, economic and physical systems, as well as recognizing their inter-dependencies to develop comprehensive solutions. This paper presents the findings and recommendations of a Team Project from the International Space University's (ISU) Space Studies Program 2025 held in the Republic of Korea, which involved interdisciplinary collaboration among an international group of participants. The project explores how satellite data can be used for designing adaptation and resilience strategies for making decisions that enable societies to be more sustainable and prepared for climate change.

This research proposes a comprehensive strategy for effectively integrating space-based data and technologies to mitigate vulnerability and strengthen resilience against the impacts of climate change. Case studies are considered, such as coping with urban heatwaves, flooding, landslides and wildfires induced by global warming and rising temperatures. This interdisciplinary framework recognizes the interconnected nature of the impacts of climate change on our society, including disaster resilience, health and well-being, nature and biodiversity, built infrastructure, and economic sectors. Satellite data provides important tools for monitoring and tracking key indicators of climate change, while playing a key role in developing early warning systems, mitigating risks, and establishing community adaptation and resilience strategies. Leveraging satellites and space-based data is critical to enhancing society's ability to respond to climate change and ensuring sustainable development for the future.

Keywords: *Climate change; Flooding; Heat Islands; Global Warming; Earth Observation; Satellite Applications*

1. Introduction

Scientists are convinced that the impacts of climate change are unevenly distributed. Certain regions are experiencing a high concentration of hazards and an elevated risk of loss of life. This project contains case studies into two of these regions, namely the Republic of Rwanda and the Republic of Korea. The studies will focus on the consequences and related events of floodings, landslides, Urban Heat Islands (UHI). Forementioned geographical areas are becoming a priority to decision makers and international organizations, with some industrialized and technologically advanced countries accepting their part of the responsibility in supporting developing nations (United Nations Framework Convention on Climate Change). Most adaptation and resilience efforts have been top-down,

considering the needs of the local communities that are affected by the consequences of climate change. This paper will draw on existing technologies that help nations to tackle the climate crisis through a data-driven response at regional and community levels. The use of satellite-based solutions will drive this paper through the two case studies. Both case studies will focus on the general issues of climate change, to then deep dive into the detailed scenario and event analysis regarding their individual situation. In this overview paper, the community-based satellite data applications present how the combination of Remote Sensing, ground-based measurements, GNSS, demographic and GIS, and advanced Artificial Intelligence (AI) techniques for data analysis and visualization (S.Lee *et al.*, 2015) help public and private organizations technological capacity and resources readiness towards disaster

risk reduction. In supporting the transition from vulnerability to resilience, this paper contributes to the academic literature by advancing the use of satellite data in addressing climate change. Spanning both rural and urban contexts, this paper anchors its analysis on key technological capabilities to mitigate floodings, landslides and the UHI phenomenon. From a technical perspective, this study uses an integrated approach to illustrate the value of Earth Observation satellites for the evaluation of UHI's mitigation and the value of Synthetic Aperture Radar satellites for mitigating floods and landslides. This study will include, among others, the broader framework established by the United Nations Framework Convention on Climate Change, the International Space and Major Disaster charter, the UN Sustainable Development Goals. An actionable pathway will be offered with practical solutions and tailored recommendations to support stakeholder dialogue across a broad spectrum of programmatic, technical, capacity building opportunities and new technology introduction regarding climate change resilience and adaptation.

2. Satellite technologies for climate change

This investigation considers how, with the help of space-based satellites and data, innovation, policy, and technology can shape a more secure, resilient and adaptable climate future. These rationales will be demonstrated through the analysis of the existing satellite-based technologies. Thus, this section will elaborate on what the existing technologies are that can be matched to floodings, landslides and UHI issues.

Satellite Remote Sensing (SRS): Satellite remote sensing technology has ushered in a new era of Earth observations, providing multi-scale information on Essential Climate Variables independent from in situ measurements and models simulations (Morris, 2005). This technical capability deepens our understanding of climate change from a space-based perspective and informs policy-making efforts to address it (Yoo, 2022). SRS enables multiscale observations of the atmosphere, Earth's surface and subsurface using sensors that do not require direct contact with the target. High-resolution Earth observation imagery enables detailed mapping of land cover, urban morphology - including building heights, wall areas and paved surfaces - and the distribution of vegetation. However, this kind of imagery is limited by cloud cover bias and infrequent satellite passes (EUMETSAT, 2020). One proposed solution to these challenges in the following study is the use of next-generation all-weather systems that combine microwave and thermal sensors, offering the capability for continuous monitoring.

Ground-Based Measurements: Ground-based data collection complements satellite observations by providing continuous measurements from meteorological stations, sensors such as Internet of Things (IoT) devices, and even airborne/drone technologies. Ground based-measurements are a powerful model that operates as a complementary solution to the continuous satellites data links. In Seoul, the Smart Seoul Data of Things (S-DoT) network consists of more than 1100 compact sensors installed in diverse environments such as neighborhoods, riversides, and mountain areas to record variables like (i.g air temperature, humidity, illumination, and ultrafine particles) every two minutes (Seoul Metropolitan Government, 2021). These IoT sensors complement the denser Automatic Weather Station (AWS) and Automated Synoptic Observing System (ASOS) networks, which provide standardized observations of air temperature, wind speed, humidity and radiation across the metropolitan area (M.-S. Park & Baek, 2023).

Geographic Information Systems (GIS): constitute advanced computational frameworks designed to acquire spatially referenced data from diverse sources, transform such data into analytically usable formats and facilitate its storage, retrieval and manipulation for subsequent analysis. These systems are further capable of generating customized outputs aligned with a specific demand. One of the principal strengths of GIS lies in its capacity to train an extensive, multilayered and heterogenous spatial database, while enabling queries regarding the existence, spatial distribution and attributes of a broad spectrum of geographic entities (Seo & Park, 2017).

Global Navigation System (GNSS): Although GNSS use in the climate domain is limited, techniques such as GNSS radio occultation (RO) and GNSS reflectometry (GNSS-R) can be used to strengthen the meteorological context for UHI detection. It is important to highlight the contribution of Radio Occultation, which provides vertically resolved temperature and humidity profiles, as well as GNSS-R, which has the function of informing the surface conditions. Moreover, the European Galileo (European GNSS) system enhances these inputs via higher signal availability and accuracy, delivering faster, more frequent updates that enable timely tasks, forecasting and early warning predictions (EUSPA, 2024). Equally important for UHI detection, GNSS provides (almost) precise locations and timing for all ground measurements used to calibrate and validate satellite data on Land Surface Temperatures (LST).

Artificial Intelligence Implementation: The current maturity of AI research and applications, positions the advanced computer systems as highly suitable for space-based technologies. A growing interest in commercial and ground-based operations are clearly visible. While numerous AI implementations have been demonstrated in commercial and ground-based operations, there is a growing interest in deploying AI directly onto space platforms, with promising applications such as image processing and anomaly detection (a.i. solutions, Inc., 2025). Among the algorithms employed, methods such as Random Forest have shown strong performance for classification tasks (i.g : in image recognition), and in some cases tasks applied to continuous outcomes like temperature prediction, though they are less suited for sequential time series forecasting where approaches such as Long short-term memory model network (LSTM) are more appropriate (Salman et al., 2024). However, there are numerous challenges with the integration of AI into space satellite platforms, notably regarding the hardware solutions that require sufficient memory, power generation, processing CPUs vs GPUs, algorithms training, or even modelling methods among others.

3. REPUBLIC OF RWANDA CASE STUDY

3.1: Overview of Climate change in Rwanda

Climate Change intensifies the global hydrological cycle, increasing the likelihood of short, intense rainfall. Air can hold about 7% more water vapor per 1°C of warming as described by the Clausius–Clapeyron relationship, which is a thermodynamic principle linking temperature to the atmosphere’s moisture capacity (Trenberth et al., 2015). The AR6 (The Intergovernmental Panel on Climate Change Sixth Assessment Report 2021) estimates a 4–8% rise in extreme-precipitation intensity per degree, especially in the tropics (Seneviratne & Zhang, 2021). Attribution studies show that 74% of extreme events are now more likely to occur or are more intense due to human influence, with over 60% of heavy rainfall and flood events showing a clear anthropogenic signal (McSweeney & Tandon, 2024). In East Africa, the March–May 2024 long rains were about twice as likely and 5% more intense from warming, amplified by urbanization (World Weather Attribution, 2024). Rwanda relies on hydropower for half of its electricity, making it even more vulnerable to changes in rainfall. Studies for Rwanda and adjacent basins report increases in rainfall extremes since the 1980s and project further intensification under continued warming, flash-flood risk, and slope instability in the highland districts (N. J. Sebaziga et al., 2025). Rwanda’s high elevation

provides a generally cool, temperate climate with higher rainfall in the west. This can be visualized using a Köppen-Geiger classification map as presented in the final team project report of this case study.

Climate change has triggered irregularities of rainy seasons in Rwanda with intensification of short but heavy rains. Combined with a unique topography of the country characterized by high mountains, such heavy rains lead to landslides and flooding. Rwanda is not only impacted in terms of human losses but the economy of the country is also affected. The use of satellite technology can help in mitigating the impact. To demonstrate it, this case study will cover the scientific basis of climate change in Rwanda; the link between climate change and local issues, the existing satellite solutions and its implementation status, along with the general analysis of the existing policy and international financing frameworks to support Rwanda.

3.2: Flooding and Landslides in Rwanda: Scientific and Socio-economic analysis

Rwanda is a landlocked country in East Africa. It is one of the most densely populated countries in Africa with a population of 14.2 million over a surface area of 26,798 km² (Data, 2024). It is situated in a highly elevated region, with altitudes ranging from 950 - 4500 m. Rwanda has a tropical climate, with two wet seasons (March - May, September - December) and two dry seasons (January - February, June - August) (Karamage *et al.*, 2016). Despite minimal contribution to global greenhouse gas emissions, amounting to 0.01 % of global greenhouse gas emissions in 2021 (Boyle, 2024), Rwanda is significantly affected by the consequences of climate change. In recent decades it has experienced shifts in rainfall patterns, increased temperatures and a rise in the frequency and intensity of extreme weather events. These extreme weather events can result in severe floodings and landslides that are detrimental to the local population, resulting in damaged infrastructure and death. For example, the May 2023 floods and landslides, resulted in over 130 lives lost in Rwanda, (Samba.C and Macaulay.C, 2023). Between 1980 and 2011, floods affected more than 5.5 million people and caused direct economic losses of more than USD 105 billion (Republic of Rwanda, 2022). It is projected that climate change will push 130 million people into poverty over the next decade, creating multiple development setbacks (Balasundaram, 2023). Even though there are challenges yet to be solved in the procurement and usage of impactful data, the use of space-based technologies shows promise for mitigating against climate-change-induced disasters.

This case study looks into the situation of landslides and flooding in Rwanda, investigates the current use of satellite technology for the management of these hazards, and provides some recommendations.

3.3: Flooding and Landslides in Rwanda: methods and drivers and Social Vulnerability

In climate science, the term “hotspot” can refer to different dimensions of risk. Climate warming hotspots are regions that warm faster than the global average, which can amplify atmospheric instability and extreme weather. Whereas vulnerability hotspots are areas where the combination of hazard exposure, high population density and limited adaptive capacity leads to elevated climate risk (Giorgi, 2006; JRC European Commission, 2023). Rwanda represents a vulnerability hotspot. The University of Notre Dame ranks Rwanda as the 19th most vulnerable country to climate change (University of Notre Dame, 2025). Besides, Warmer near-surface temperatures in hotspot regions can raise Convective Available Potential Energy (CAPE), a key measure of instability that influences the likelihood of deep convection and severe weather (Agard & Emanuel, 2017). Higher CAPE values are associated with stronger updrafts, more condensation and higher rainfall rates. This data is particularly relevant for mesoscale convective systems (MCSs), which are large storm complexes that contribute significantly to extreme rainfall events (Schumacher & Rasmussen, 2020). In the case of Rwanda, cities generate their own heat. Kigali, for instance, has seen a rise in LST over the past 30 years, driven by rapid urban growth. Expanding built-up areas, combined with the loss of vegetation, have intensified the urban heat island (UHI) effect (Safari Kagabo et al., 2024).

It is to be noted that the positive feedback loop method can be a powerful tool for understanding floods and landslides. In scientific jargon, warming and local heat hotspots load the near-surface air with moisture and raise atmospheric instability. Flooding leads to increased impervious surfaces and soil compaction, thereby reducing water infiltration and exacerbating runoff. Loss of natural “cooling” increases the local heating and runoff in the next storm, further boosting instability and peak flow events (Seneviratne & Zhang, 2021; World Bank Climate Change Knowledge Portal, 2020). Geographic hotspots and anthropogenic hotspots in Kigali interact with steep topography and moisture flows to produce some of East Africa’s most important and intense short-duration rainfall events. Addressing Rwanda’s flood risk will require integrating hotspots and landslides susceptibility maps into adaptation planning to identify high-risks zones. While Rwanda’s national landslide

susceptibility maps offer a broad overview of risk, they are limited by the precision of their data. To answer this challenge, this team project analyzed the possible existing satellite based data, and some key indicators that this report intended to present: an “Integrated Satellite-Based Resilience Framework”.

3.4: Integrated Satellite-based Resilience Framework for monitoring Floods and landslides in Rwanda

When used purposefully, satellite-based data collection can prove invaluable for scientists and policy-makers by providing accurate, reliable data across wide regions. In the case of Rwanda with its remote and mountainous terrain, satellite platforms are likely to be the most efficient way for taking measurements. Around the world, tools such as INFORM Risk Index, Flood Resilience Index (FRI) and Recovery Resilience Index (RRI) have strengthened disaster management in contexts ranging from humanitarian risk assessment to urban flood recovery (JRC European Commission, 2023; (Keating et al., 2017). In Rwanda, United Nations Satellite Centre (UNOSAT) operational flood maps, and national damage assessments already exist but remain unintegrated. It is in this context that this study explored the possibility of integrating INFORM solution for early risk mapping, FRI for a real-time impact assessment, and RRI for a post event recovery solution. As a result, these combined frameworks shall cover the entire disaster lifecycle. Such an approach would provide faster, more targeted and locally tailored disaster management, aligned with the Sendai Framework priorities (United Nations, 2019).

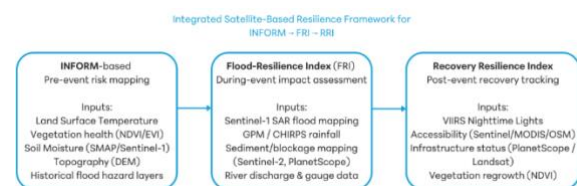


Figure 1: Integrated satellite based resilience framework for Rwanda.

The initiative of the Rwandan Government to reduce flooding and landslides through a better prediction is relevant due to the direct socio-economic impacts. Disasters in Rwanda cause significant economic losses, costing the country approximately 1% of its Gross Domestic Product (GDP) annually from landslides and 5% from floods (World Bank, 2024). The financial toll amounts to around USD 200 million per year in damages due to landslides (Rwanda Climate Change, 2024). However, this figure surged dramatically following the devastating floods in May 2023, with losses rising to USD 415

million, highlighting the increasing economic impact of extreme weather events on the nation (Minecofin, 2024).

Flooding and landslides not only destroy livelihoods, they also erode the soil that sustains crops. The agricultural-based economy employs 65% of the Rwandan population, and, as a result, usable land is eroded and degraded to meet these agricultural needs (Balasundaram, 2023). As mentioned above, satellites offer an important tool for mitigating and adapting to the impacts of flooding and landslides, they are known for their cost effectiveness and technological solutions. This study has revealed four relevant satellite solutions for flooding and landslides: Earth Observation (EO) satellites combined with GIS, telecommunications satellites and GNSS. GNSS data has the possibility to be geotagged with the current EO data. Besides, data fusion between multiple solutions could also bring significant input, such as the use of Synthetic Aperture Radar (SAR) that provides visibility of the ground through clouds, smoke, rain at night, but is difficult to interpret (J. Park et al., 2024; Schumann et al., 2018). In addition, before floods, there are risk mapping and simulation tools that require accurate Digital Elevation Models (DEMs) and hydrological models. During floods, satellites enable Early Warning Systems (EWS), flood extent mapping, evacuation planning and relief effort coordination. Post flood, satellites are used for the recovery and restoration efforts to identify and treat erosion, sedimentation and land changes. For data processed through the GIS model, the study demonstrates how this solution is practical for Rwanda. Many papers have been written on its effectiveness. One meaningful example is the use of GIS data to predict landslides susceptibility that compares the performance of multiple modeling methods using machine learning (J. B. Nsengiyumva & Valentino, 2020). In the same trend, the Rwanda Space Agency is incorporating satellite data for the management and prediction of landslides and floods using its GeoSpatial Data Hub (GEOHUB), a centralized database integrating key satellite platforms and geospatial tools is under work in progress. The GEOHUB combines satellite imagery, in-situ data, demographics data and other geotagged data for modeling, and analysis using AI/ML to support the government to make data-provided decisions. The platform provides digital infrastructure to academia and startups for building innovative solutions for boosting the private sector.



Figure 2: Rwanda Geospatial HUB (GEOHUB)

The solutions hosted on GEOHUB focus on four critical areas: agriculture, disaster risk management, urban planning and environmental monitoring. In terms of flood prediction, only a limited number of applications are currently available. Few applications can process GIS data at a 30-meter resolution and incorporate multiple variables influencing flood simulations, including land cover, rainfall and infiltration. Whereas for landslide prediction tools, likewise, many resources are available to assess the landslides vulnerabilities, for instance the United States Geological Survey (USGS) provides a free online Landslide Inventory and Susceptibility Map in the United States (U.S Geological Survey, 2025). Few disaster management tools that use satellite data are specific to Rwanda. One notable effort that is currently under development is the “Geo-hydrological Hazards triggered by rain in Tropical Africa: a demonstrator for Rwanda to document the effects of Climate Change” (GeoHaTACC). GeoHaTACC will use data from Sentinel-1 and -2, Landsat-8/9, and Planet Scope, as well as other precipitation data to detect and inventory hydro-geological hazards (A. Deijns, 2022). GeoHaTACC is a collaboration between Rwanda and European entities, including CNES (the French Space Agency) and Centre national de la recherche scientifique (CNRS, the French National Research Center). One of the major concerns for both flooding and landslide applications is the data processing per local community. To create the possibility of data forecast, the prediction model shall be trained with a high level of accuracy, but some areas may have limited access to radar observation needed to create such high-fidelity forecasts. The project goal is to create automatically produced geohydrological hazard inventories. The pixel level accuracy allowed the developing team to supervise a classification method that helped using machine learning methods to make a territory screening for decision making purposes. The outcomes of the systematic, regional detection using Sentinel-2 were tested in countries other than Rwanda; the outcomes showcased its effectiveness. The scaled-up calculations were made possible through the collaboration between the project and the Data-Terra research infrastructure and via its THEIA and ForM@Ter thematic Cluster (Space for Climate Observatory, 2022). Several projects were

also initiated in 2023 through the implementation of the Green Climate Fund. The fund supported an international contribution of USD 42.8 million to the Ireme Invest, an investment facility that works with Rwanda's private sector to support green business growth to boost the country's response to climate change (Ireme Invest, 2024).

In addition to the current successful projects that are ongoing, it is to be noted that the Rwandan Space Agency (RSA) is proactively working on the development of an early warning system. The system consists of a model that combines the soil composition and a set of topographic data, soil moisture, and land cover highlighting areas that are likely to be affected by landslides and floodings, depending on the level of precipitation. The system has the capability to automatically send SMSs to the right people for quick intervention. To support such initiatives, there are a few non-negligible technologies that can help the Government of Rwanda in planning a community centered based technology.

Now that we have set the stage on a generic presentation of the current technological possible leverages on the GIS, it is necessary to deep dive into the details to understand other generic technologies. Satellite-based solutions are deployable to improve connectivity, particularly in remote regions where ground based communication equipment is not often available or even implementable due to the terrain's quality and the costs of constructing appropriate infrastructure. In Rwanda, mobile penetration is already at a good percentage, with 85% of households owning at least one mobile phone in 2024, including 34% equipped with a smartphone. Furthermore, the government's school Connectivity Program, in collaboration with Starlink that aims to connect 500 schools in its initial phase - of which 50 are already connected, serving over 18,000 students. Commercial operators using satellite constellations in LEO, such as Starlink and OneWeb (and in the future IRIS2), allow for internet access in remote locations around the world. A key technological advancement in this context is Direct-to-Cell (D2C) satellite connectivity, which enables ordinary mobile phones to connect directly to satellites in low Earth orbit (LEO) or geostationary orbit (GEO), bypassing damaged or overloaded terrestrial infrastructure. This provides continuity of communication without requiring special satellite phones or ground relays. Because the connection is satellite based, it covers wide areas, including remote or isolated. Besides, Hybrid Satellite Ground Systems, combining GSM, LoRa further strengthen resilience in disaster management.

To conclude with this primary case study: we noted that Rwanda's commitments to the climate change

resilience and adaptation were translated by the implementation of the Environment Policy, revised in 2015, which gives it the necessary level of readiness to strengthen its environment and climate change governance. Through the first National Disaster Risk Reduction and Management Policy (Government of Rwanda, 2023) updated in 2022, the Republic of Rwanda intends to align the national initiatives towards the international goals. It should be noted that the local government is supported by the Information and Communication Technologies (ICT) national framework. Leading the RSA, the ICT priorities is also emphasized by the Digital Inclusion through accessible, resilient, secured Infrastructure. Despite the modernization and expansion of access to communication and internet infrastructure; the development of high-performance infrastructure; the programmatic vision of a national communication and space infrastructure for sovereign capabilities; or the improvement of the quality of service and experience through mobile number portability and Converged Network Management System could be part of the field experience feedback background for this case study. To achieve this ambition, Rwanda can count on the UNFCCC funds (United Nations Framework Convention on Climate Change, 2023). In addition, the Agence Française de Développement (AFD), is also partnering them through a number of projects. Space for Climate, Connect by Cnes investment initiatives, the Copernicus Programme, the German Aerospace Center, or the Japan Aerospace exploration Agency are several nodes where the Rwanda Government shall obtain support for their future endeavors.

4. REPUBLIC OF KOREA CASE STUDY

4.1: Overview of Climate change in South-Korea

Since the 1970s, satellite observation data have played a pivotal role in monitoring global climate change and environmental conditions. In the Republic of Korea, research on satellite-based meteorological observation began in the 1980s, initially relying on foreign satellite data for weather forecasting. A major leap occurred in the 2000s with the launch of the nation's first geostationary meteorological platform, the Communication, Ocean and Meteorological Satellite (COMS). The launch of COMS also enabled the Republic of Korea's capability for independent satellite data processing. The subsequent deployment of the Geo-KOMPSAT-2A (GK2A) brought notable enhancements in spatial, temporal, and spectral resolution, thereby expanding the range and precision of the Republic of Korea's satellite-derived

products (Chung et al., 2024). More recently, the integration of artificial intelligence and advanced data analysis techniques has further extended the utility of satellite observations, particularly in nowcasting, forecasting, numerical weather prediction and climate change monitoring (Oh et al., 2020). In 2018 the Republic of Korea committed to reducing greenhouse gas emissions by 40% by 2030, according to the Nationally Determined Contribution (NDC) of the Republic of Korea to the Paris Agreement (United Nations Framework Convention on Climate Change, 2021). Currently, Earth experiences an energy imbalance of $\sim 3\%$, resulting in net warming of the Earth (Loeb et al., 2021). While greenhouse gases are essential for maintaining habitable temperatures, their uncontrolled anthropogenic emissions intensify the effect of global warming to a greater extent (NASA, 2021). It is estimated that urban areas account for around 75% of global energy consumption and 70% of global greenhouse gas emissions (International Energy Agency, 2024). With a high concentration of human activities in cities and urban areas, such as transportation, energy use, industrial processes and construction it is a major source of greenhouse gas emissions. Subsequently, there is also a positive feedback mechanism in action. Greenhouse gas emissions intensify UHIs through global warming and altered urban microclimates. Not to mention the thermal and radiative properties that influence the UHI phenomenon.

4.2: UHI's in the Republic of Korea: Scientific and Socio-economic analysis

UHI's are considered as local phenomena, which are fueled by human activities, especially the burning of fossil fuels and land-use changes. Climate change raises global temperatures, while UHI's cause localized warming in urban areas. Following a thorough review of literature, the key factors which affect UHI's are:

- **Surface Materials:** Asphalt, concrete and dark roofs absorb and retain heat.
- **Urban Geometry:** Tall buildings and narrow streets create "urban canyons" that trap heat and reduce airflow.
- **Building Density:** Dense development limits green space and increases heat retention.
- **Human factors:** Pollution, vehicle emissions, and heat emissions from buildings

Over the past century, the Republic of Korea's national average temperature has increased by approximately 1.8°C , surpassing the global mean of 1.1°C for the same period (Forster et al., 2025). Precisely, in Seoul, this temperature increase has

been intensified by rapid urbanization and the urban heat island effect. The results of such have been marked by an extreme heat event, including more frequent, prolonged and intense heatwaves, as well as a reduction in night-time cooling. Extreme urban heat mitigation approaches are not only a real environmental hazard, but also a socio-cultural and economic challenge. Social vulnerability can refer to the socioeconomic and demographic factors that affect the resilience of communities (Flanagan et al., 2011). The author developed a social vulnerability index (SVI) to mitigating, preparing for, responding to and recovering from disaster related risk. Thanks to recent technological returns from advancement in big data analytics through mobile devices, continuous high-resolution insights are generated for the purpose of ensuring adaptation measures both to protect those most vulnerable and reflect the lived realities of local communities. Besides, studies show a strong link between artefacts such as urbanization and vulnerability to heat-related mortality in the Republic of Korea (E.S Kim et al., 2025). The topic on health implications of urban heat are well documented in the Republic of Korea contexts. In a national and district level time-series study it was found that urbanization significantly amplifies vulnerability to heat-related mortality. Similarly, (S. Kim et al., 2025) analyzed the nation's capital region, the results demonstrated that urban heat anomalies, quantified through LST, are strongly correlated with excess mortality during heat events, especially in densely populated districts with low Normalized Difference Vegetation Index (NDVI) score.

After the overall examination of causes and features of the UHI, the next section of this paper will discuss the methods, drivers and space-based tools that can be used to mitigate UHI effects.

4.3: UHI's in the Republic of Korea: methods and drivers and Social Vulnerability

To identify, implement and monitor the right solution for UHI reduction, it is necessary to list the existing key parameters that are used to measure and analyze the UHI using satellite data. The most direct indicators include thermal parameters such as LST. This data is often derived from thermal infrared sensors on satellites such as the Moderate Resolution Imaging Spectro radiometer (MODIS), Landsat and Sentinel. Both day and night time LST is important when studying UHI effects, as cities often stay warmer at night than rural areas. Another parameter that can be cited is the Surface UHI (SUHI) intensity, which is calculated as the temperature difference between urban areas and the surrounding rural areas. The Normalized Built-up Index (NDBI) identifies the built-up areas of buildings and constitutes the

bulk of concrete structures; this indicator specially helps to distinguish urban zones from natural ones. Finally, a parameter of significant importance is the albedo, or reflectivity of the surface. Additionally, carbon emissions data from ground measurements can also be an integrated part of the environmental impact assessment parameter. To complete the parameter analysis, a few examples of sensors types that are mobilized are as follows

Optical/Visible	Land cover and land use changes, vegetation health, albedo changes, surface bodies of water and post-wildfire scars (Liao <i>et al.</i> ,2022)
Thermal	Land surface temperature, evapotranspiration, and soil moisture (McCartney, 2023)
Lidar	Tree canopy height, ice sheet and glacier elevation changes, and vertical cloud profiles (Xu <i>et al.</i> ,2023)

Table 1: Sensor types to monitor UHI effects

In addition, meteorological networks and in situ measurements provide complementary ground-level data for studying UHI. However, each of the above-mentioned methods has their limitations: satellite data often lacks the accurate spatial resolution to evaluate human-experienced temperature, while ground-based measurements are limited in geographic scope and scalability (Sobrino *et al.*, 2021). To address the limitations, (Shandas *et al.*, 2019) a novel framework has been developed by integrating Sentinel-2 satellite imagery with ground-based temperature data collected via mobile traverses. Using Random Forest regression models, the study results demonstrated few temperature differences across Richmond, Baltimore and Washington, D.C., which revealed the strong correlation between land use and land cover features. As a result, it has been highlighted that the integration of spectral satellite data with in situ observations produced highly predictive models with low error margins, offering a cost-effective approach to urban heat assessment. To illustrate the UHI drivers and cooling effects methods, researchers have demonstrated that some vegetation-rich areas, particularly urban forests, became an important part to reduce UHI risks. A contrast is particularly visible between areas where there is a higher urban density and less urban vegetation versus areas where urban vegetation (such as green corridors, rooftop gardens among other initiatives) helped reduce land surface heat. Urban forests in Seoul recorded an average cooling of approximately 1.9°C over distances of up to 300m from the forest edge (B. Hwang *et al.*, 2023). These findings

reinforce the importance of integrating remote sensing data with ecological and urban design strategies for thermal resilience.

Despite the various UHI assessment mechanisms, several authors recognize the limitations of a static approach. In that approach, (Yoo *et al.*,2020), proposed a diurnal heat risk assessment framework that integrates high-resolution air temperature data, modeled using a stacking ensemble machine learning algorithm, to leverage the strengths of different models to improve accuracy, with a real-time data and pre-determined socioeconomic indicators. In this same context, (K.Park *et al.*, 2023) further observed the interaction between UHI's and heat waves using 25 years of observational data. These authors found that UHI intensity increases during heat waves by 0.53°C at night and 0.20°C during the day, demonstrating a feedback effect that exacerbates thermal stress. These interactions vary depending on prevailing concise patterns: The Pacific-Japan (P-PJ) pattern that amplifies the UHI effects and has become more frequent in recent years, whereas the expanded western North Pacific subtropical high (E-WNPSH) pattern moderates UHI intensity through higher humidity and cloud cover (K.Pak *et al.*, 2023). These results suggest the need to integrate atmospheric dynamics into urban heat risk models. Narrowing down the study to the Republic of Korea, (Priyankara *et al.*, 2019) conducted a geospatial analysis of SUHI formation in the Seoul Metropolitan Area using Landsat data from 1996, 2006 and 2017. The findings bring suggestions on how urban planners can prioritize green infrastructure and vertical development to mitigate SUHI effects.

Expanding the analysis of the UHI effects through a business lens, urban heat presents a multifaceted challenge for businesses in the Republic of Korea, influencing operational costs, workforce productivity, asset valuation and others. The impact of urban heat on business is mediated by geographic, socio-economic and infrastructural factors, particularly urban density and vegetation coverage, directly influencing localized temperature extremes (E.S. Kim *et al.*,2025; Son *et al.*, 2016). Research shows that population density and NDVI are valuable indicators of business risk in urban areas (E. S. Kim *et al.*, 2025; Son *et al.*, 2016).

Flanagan *et al.* propose a Social Vulnerability Index containing fifteen metrics for disaster management that are widely cited in recent environmental analysis, medical research and disaster management policy (Flanagan *et al.*, 2011b). In a second study, (D.-W. Kim *et al.*, 2017) analyzes the spatial correlation coefficients of eleven metrics specific to heat vulnerability in the Republic of Korea, which are also cited in recent heatwave vulnerability

studies across the world. Kim uses exposure, sensitivity, and adaptive capacity to categorize the eleven metrics, following the widely recognized climate vulnerability index framework. (D.W. Kim et al., 2017), down-select the eleven metrics (i.g: income, poverty, employment, education, age, single parenting, disability, language, elderly living alone, agricultural workers and tropical nights) keeping the strongest correlation from each to form a Heat Vulnerability Index (HVI). The study states that there is an approximately linear relationship between the number of heatwave-related deaths and the HVI. However, the R-squared value, indicating the proportion of variance in one variable that can be explained by the other, is 0.32, which lowers the confidence of their conclusion and suggests that a better index could be proposed to describe heat vulnerability. In a third study, Lee and Han focus on the effectiveness of cooling center (typically air-conditioned) distribution in Seoul (J. S. Lee & Han, 2024). They use the methodology from D.-W. Kim et al. (2017) of combining exposure, sensitivity and adaptive capacity metrics. Furthermore, other studies suggest that access to medical infrastructure is as important as social isolation, hence the other adaptation metrics from the result of this study should be included in an improved index for further analysis. Having analyzed the socio-economic, scientific methods and drivers used to evaluate urban heat island effects in the Republic of Korea, the discussion moves to the current strategies for UHI monitoring and space-based technologies that serve as monitoring instruments to climate-change related variables.

4.4: Integrative Infrastructure for Monitoring UHI's

The detection and monitoring of urban heat islands relies on an integrated approach that can combine Satellite Remote Sensing, ground-based systems, GNSS, GIS, and AI techniques for data collection and analysis (S. Lee et al., 2015). Over the past decade, advances in artificial intelligence have significantly strengthened the integration of diverse data sources for Urban Heat Islands analysis. And, ground-based networks such as the Korea Meteorological Administration's 77 manned meteorological stations and 464 weather stations provide high resolution temporal monitoring that complements, rather than competes with, satellite-based observations, enabling more accurate and periodic assessment of urban heat dynamics (Xiong & He, 2022). A central focus within this framework is on data fusion, ensuring that multi-source observations are coherently combined, and that the resulting insights can be effectively distributed

across the different operational departments of the End User.

4.5: Satellite Remote Sensing (SRS) technology:

SRS is able to monitor a wide range of climate change-related variables including clouds, aerosols, greenhouse gases, land use and land cover changes, glacier ablation, water and energy fluxes, and sea level variations (Matson et al., 1978). Optical and thermal infrared satellite datasets such as Landsat-8, MODIS, ECOSTRESS, and Sentinel-3 are used to monitor LST and identify UHI patterns over time (Price, 1979). The national space capabilities of the Republic of Korea help the country to evaluate and monitor UHI's within a multi-layered observation system that integrates domestic space-based assets with international satellite missions and high resolution airborne and ground-based measurements. While the Republic of Korea operates its own earth observation and remote sensing satellites, coverage is further strengthened by data from global platforms such as NASA's Landsat program and the MODIS. This integrated framework enables the monitoring of heat phenomena at scales ranging from continual to neighborhood level, supporting research on environmental change, public health impacts and the development of targeted mitigation strategies (Chung et al., 2024). Besides, after a meticulous study, it is considered that KOMPSAT-3A is particularly valuable for UHI studies thanks to its Mid-Infrared/Thermal Infrared (MIR/TIR) sensor, offering 3.8m resolution thermal data that enables precise mapping of heat storage and wildfire detection among others. This study examines the UHI effect in Seoul by comparing summer conditions in 2014 and 2024. Using satellite remote sensing, the analysis provides a consistent, city-wide view of how surface temperatures and land cover characteristics have evolved over a decade. Two main components form the basis of the work. The first focuses on temperature-related metrics: LST and two derived indicators: UHI intensity and the Urban Thermal Field Variance Index (UTFVI). These indicators provide both absolute and relative measures of thermal patterns across the city. The second examines land cover features closely linked to urban heat, using the NDVI to measure vegetation cover and the NDBI to identify dense urban areas. By combining thermal indicators with land cover indices, and breaking the analysis down by district, the study aims to reveal not only where heat is concentrated, but also why certain areas are more affected, offering insight for future heat mitigation efforts. For the temperature study, band 10 has been used to provide the thermal information needed to estimate surface heat patterns across the city. For the

land cover study, bands 4, 5, and 6 have been selected: band 4 captures vegetation in the red spectrum, band 5 highlights greenery in the near infrared, and band 6 helps distinguish built-up areas.

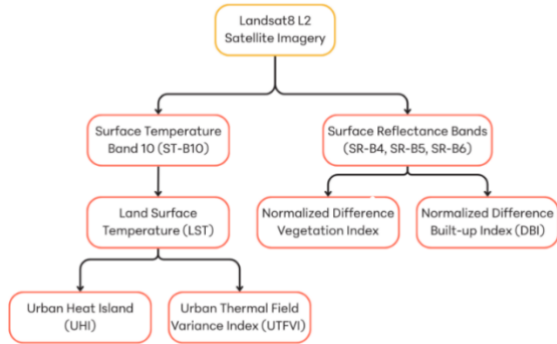


Figure 3: Computation Flow Chart

Together, these bands allow the comparison of vegetation and urban density with temperature distribution. The decision to use Landsat imagery, specifically Landsat-8, over domestic KOMPSAT or others for analysis of UHI's across the Republic of Korea's districts is based on data accessibility, operational feasibility, and data suitability. Besides, in this study the LST was derived from Landsat-8 Level-2, Collection 2, Tier 1 imagery for the summers of 2014 and 2024. The workflow was carried out in Google Earth Engine (GEE) and the final datasets were exported for mapping Quantum Geographic Information System (QGIS). These data export methodology details can be found in the final report of this team project work. Beyond the first set of results obtained from a pattern range of values for each year, an attempt was made to compare the mean temperature observed in the ground observatory and satellite data. It is also to be noted that despite the basic calculation method used from UHI intensity by comparing LST values against a reference baseline and UTFVI, the comparison ensured consistency and classification and legend scaling method across years. Traditionally, the UHI calculation falls into three categories: first, the direct calculation method based on LST, the second based on urban-rural temperature differences, and finally, inferential methods based on temperature while controlling for other environmental variables. To analyze the 25 administrative districts in Seoul for this study, we applied the Simplified Urban-Extent (SUE) algorithm, which was proposed by Chakraborty and Lee (2019). The core idea of this algorithm is to calculate the difference between the LST of urban pixels and the study area's average LST using Eq.

(1).

Eq.(1)

Following the UHI, the UTFVI value was also used to evaluate the comfort level of the urban thermal environment to determine the urban health condition using Eq. (2).

Eq.(2)

$$UTFVI = \frac{LST^{pixel} - LST^{mean}}{LST^{mean}}$$



UHI and UTFVI maps for each administrative district of Seoul were obtained using Landsat-8 satellite data. UTFVI has become an increasingly commonly used indicator to more accurately describe the SUHI effect (Kafy et al., 2021). Besides the UHI, UTFVI key measurement index, urban vegetation, impervious surfaces, and bodies of water also play a key role in how heat is absorbed, retained, and released within the city. To capture these factors, two widely used spectral indices were calculated from the surface reflectance bands of Landsat-8. As mentioned, the NDVI measures vegetation density and greenness whereas the NDBI helps in detecting built-up impervious surfaces.

$$Eq.(3). \quad NDVI = \frac{NIR - RED}{NIR + RED}$$

As mentioned above, 25 official administrative districts were analyzed. By using a detailed map file consisting of LST, UHI intensity, the Urban Thermal Field

Variance Index $UHI = \frac{LST^{pixel} - LST^{mean}}{LST^{mean}}$ Index (UTFVI), and three land cover measures (NDVI; NDBI, NDWI) were gathered. A general comparison with the ground data was carried out and it was found that the satellite data was 11 to 12°C higher than the ground data. The LST increase was 0.8°C between 2014 to 2024, and the temperature increase around 1/1°C for the same period of time. The coverage ratio also allows measuring the quality accuracy of the data according to the expected requirements. District wise change in LST and NDVI from 2024 to 2014 is shown in Figure 20 below. Compared to the nominal average LST increase of 0.96 °C, cities like gangseo-gu show a maximum increase of 3.5 °C. In many districts, when NDVI increased, LST decreased, which implies higher vegetation is helping to cool the surface. Some districts show both NDVI and LST increasing or decreasing together, which implies inconsistent correlation between the two variables. Overall, the

change in average NDVI index during this period shows a weak correlation with LST (-0.17). Seasonal variation in vegetative index is one factor, as the data considered is only a three month average (for June, July, and August) between 2014 and 2024. Similar weak negative correlation has also been reported in other studies (Guha & Govil, 2020). The magnitude and direction of change vary significantly across districts, highlighting the role of local urban planning, land use, and microclimates. In this study, the normalized difference built-up index ΔNDBI remains relatively consistent across all districts. Despite this stability, changes in LST vary greatly, with some districts experiencing notable warming and others cooling. It is also to be noted that this study has also taken into account the comparative distribution of population by census across Seoul. The results show the districts of Songpa-gu (with a population of 628,648 people), Gangseo-gu (543,974 people) and Gangnam-gu (528,194 people) as the most populous areas in Seoul in 2024 (“KOSIS Korean Statistical Information Service”, 2019). The total population across Seoul’s 25 districts declined by approximately 5.74% between 2015 and 2024, reflecting a notable downward trend in urban residency over the nine-year period. In comparison with increased LST across the majority of 25 districts there is no strong correlation to link population to the increase in LST seen in the data. Furthermore, the data contradicts the expectation that Seoul’s most populous district, Songpa-gu, would exhibit the greatest increase in LST, given factors such as building density and anthropogenic activity. However, to better understand the influence of UHI drivers on LSTs and UHIs, further investigation is required to account for variables including district area, population density, and patterns of land and building use. Although census data does not explain the maximum LST, it is critical to understand the vulnerability metrics associated with such metrics. The Seoul Open Data Portal and SGIS can be used to compare social vulnerability metrics to the satellite-based exposure data from the previous sections. J. S. Lee and Han, 2024 improved on the analysis by D.-W. Kim et al., 2017 to define a heat vulnerability index using multiple sensitivity and adaptive capacity metrics. However the results data accuracy of the unweighted aggregation method used to calculate their index remains unknown. This uncertainty highlights a clear direction for future research such as follows :

- Integrate satellite-based exposure data with sensitivity and adaptive capacity data from the Korean Statistical Information to develop a satellite-based heat vulnerability index.

- Validate the heat vulnerability index using heatwave deaths from KOSIS.
- Apply this analysis to other urban areas in the Republic of Korea or other comparable country in the region.

The Republic of Korea’s urban heat challenge is a complex, multi-sectorial issue that intersects with climate change, public health, social equity, and business sustainability. Joint actions between policymakers and the private sector is essential. The review of the current literature, field expertise, combined with a statistical data analysis, it was possible to identify the most critical hotspots in Seoul, revealing strong overlaps with socio-economic vulnerabilities. The UHI is not only an environmental issue, but a public health and justice and equity concern. The results also confirmed that anthropogenic heat sources, such originated from transportation, industrial facilities, and widespread air-conditioning use, contribute to localized temperature peaks, creating feedback loops that increase both energy demand and emissions. From a technical perspective, the study illustrated the value of Earth observation and other space-based solutions for continuous monitoring, and evaluation of UHI mitigation strategies. Overall, this work has brought beyond simple problem identification. The outcomes on how the high-resolution spatial analysis is combined with targeted policy and community engagement, the response to climate-induced urban heat can be both scientifically robust and socially inclusive, paving the way for resilient, sustainable and livable cities in an era where climate change effects are visible.

5. Combined Recommendations and Discussions

While UHI’s in Seoul and Rwandan flooding and landslides represent two unique problems that require distinct solutions, there are common recommendations to mitigate against and adapt to these climate change induced phenomena. Collectively the recommendations in this report can be summarized into four key themes: increased adoption of satellite-based data, capacity building of new technologies, operationalization of disaster indices, and deeper collaboration with stakeholders.

5.1: Increasing the use and distribution of satellite based data

Satellite based data provides continuous, global coverage of remote areas and should be more widely integrated into current disaster tools and policies. While the Republic of Korea has a well-developed network of earth observation satellites, current data sharing policies for imagery are restrictive. More

open-data policies should be adopted to distribute satellite data more widely to stakeholders and also establish training programs to ensure users have the skills to use the data. Most critically this should include high-resolution optical and thermal infrared imagery to identify neighborhood scale UHI intensity and variance. By adopting open data policies this geotagged imagery can be combined with other sources to develop more advanced heat vulnerability products and apply Artificial-based analytics for urban heat planning, monitoring and response. Conversely, Rwanda does not currently have its own operational earth observation satellites and in the short term should focus on partnerships with regional, international and commercial providers to better leverage the existing global satellite network. In the long term the government of Rwanda should seek opportunities to deliver secondary payloads and regional partnerships for the development of sovereign satellite systems. Taking this cost-effective and phased approach, it can build domestic capacity for the processing and analysis of satellite data with its Geohub platform.

5.2: Technology Capacity Building

Technological solutions at all stages of climate phenomena should be further developed to improve prediction, response and recovery. In Seoul a digital twin of the city should integrate the large existing satellite-based datasets which can be analyzed with AI to aid in the creation of policies and decision making of national and local leaders. These models can provide guidance in the areas of integrating green infrastructure into urban planning, hot spot prediction and early intervention, and evacuation planning prior to disasters. Similarly, Rwanda should focus on improving its software tools like Geohub for the prediction of landslides and floods before they happen. Key performance parameters to improve are the latency and accuracy for early warning systems. For these solutions accurate weather prediction is critical and these models require incorporation of the unique geographic and hydrologic features of Rwanda. Technology transfer programs with developed countries can enhance the adoption of new tools such as landslide detection ground sensors. This report also presented a novel ground-based sensor solution that is enabled by satellites to provide advanced warning for landslides and floods.

5.3: Integrating operational indices

Satellite data should be made available in the form of operational disaster indices to facilitate easy decision making by key stakeholders. In Rwanda this

should include the INFORM Risk Index for early hazard and vulnerability mapping, the Flood Resilience Index for real-time impact assessment, and the Recovery Resilience Index for monitoring post-event recovery. In Seoul, a more comprehensive heat vulnerability index could be developed with additional parameters like exposure, sensitivity, and adaptive capacity, and other criteria that are tailored and adapted to the local community-based capabilities. Subsequently, incorporating multiple factors to provide a deeper understanding of the areas that are most impacted by UHIs. These indices are integrated into alert system networks that are managed in a local node.

5.4: Working with stakeholders

Disaster management requires collaboration with stakeholders at all levels and functions across government, business, academia, and communities. In Rwanda education should be a key focus to increase community openness and adaptation to emergency protocols and disaster responses. Community workshops should leverage the local network of community health outreach teams to teach as well as integrate indigenous knowledge and cultural boundaries and technology early adoption with disaster response solutions. The Republic of Korea has a well-established policy framework for UHI response, but should continue to develop sustainable business policies that address the growing issue of UHIs. This should include tax incentives and some directed governmental financial, material, training aids for businesses to upgrade infrastructures, implement key innovative reflective materials, and intensify cooling and ventilation methods to reduce heat stress effects.

6. Conclusion

By systematically reviewing the current space-based technologies in the different sectors of the economy, gaps have been identified in the current implementation strategies expressed by developing countries. Thus, the two case studies are helpful to understand the climate change effects, most notably flooding, landslides and UHI's. The report concludes that a Nature-Based Solutions (NbS) to adopt climate resilience exists, however in both case studies, a customized community-based solution or ecosystem-based adaptation initiatives may be the next step towards an integrated solution that could fit for all. Combining and mapping physical and demographic indices to quantify and visualize high-risks areas and local needs using Artificial Intelligence was one of the outcomes proposed by the team project. The data accuracy by using ground sensors in combination with satellite

telecommunications as an early alert system to vulnerable populations, and real-time data sharing for decision-making purposes is provided. Between diplomacy, policy, space-based technology evolution and international partnerships, the team project has worked on demonstrating how satellite data provide important tools for monitoring and tracking key indicators of climate change, while playing a key role in developing early warning systems, and establishing community adaptation and resilience strategies. This study has demonstrated that leveraging satellites and space-based data is critical to enhancing a society's ability to respond and how to maintain sustainable development goals as a focus for the future generation. The importance of framing data modeling for a larger trainable and replicable scope in disaster management could invite further research venues in the future.

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